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Research Scientist at Lawrence Berkeley National Laboratory (LBNL), USA. His work centers on optimizing data management and I/O performance to support scientific workflows that increasingly integrate AI. By investigating data access patterns, automatic tuning, and storage solutions, he aims to ensure massive datasets are ready for disruptive AI technologies, enabling researchers to leverage both new and historical data for scientific discovery.

EXPERIENCE

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| Lawrence Berkeley National Laboratory (LBNL, USA)
Career-Track Data Management Research Scientist | <i>March 2023 – Present</i> |
| Lawrence Berkeley National Laboratory (LBNL, USA)
Postdoctoral Scholar in Scientific Data Management | <i>May 2021 – March 2023</i> |
| Barcelona Supercomputing Center (Barcelona, Spain)
Visiting Researcher / Doctoral Stay | <i>August 2019 – February 2020</i> |
| Inria Centre de Recherche Grenoble Rhône-Alpes (Montbonnot, France)
Visiting Researcher (Datamove) | <i>January 2017 – February 2017</i> |

EDUCATION

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| Federal University of Rio Grande do Sul (UFRGS)
Doctorate in Computer Science
“Dynamic Tuning and Reconfiguration of the I/O Forwarding Layer in HPC Platforms” (Honors)
Advisor: Prof. Dr. Philippe O. A. Navaux (UFRGS)
Co-Advisor: Dr. Toni Cortes (Universitat Politècnica de Catalunya / Barcelona Supercomputing Center) | <i>February 2017 – May 2021</i> |
| Federal University of Rio Grande do Sul (UFRGS)
Master in Computer Science
“Evaluating I/O Scheduling Techniques at the Forwarding Layer and Coordinating Data Server Accesses”
Advisor: Dr. Philippe O. A. Navaux (UFRGS) | <i>February 2015 – January 2017</i> |
| Universidade Regional Integrada do Alto Uruguai e das Missões (URI)
Bachelor in Computer Science
“Implementação de Técnicas de Tolerância a Falhas no URI Online Judge”
(Applying Fault Tolerance Techniques in URI Online Judge)
Advisor: Prof. Msc. Paulo Ricardo Rodegheri (URI) | <i>February 2010 – February 2015</i> |

PUBLICATIONS

Journals

- ^{j1} Salimiparsa, A., Lewis, N., **Bez, J. L.**, Byna, S., *A 360° Survey on Reconfigurability of HPC Storage Systems for Efficient I/O.* ACM Comput. Surv. (*Accepted*)
- ^{j2} Hiniduma, K., **Bez, J. L.**, Byna, S., *Data Readiness for AI: A 360-Degree Survey.* ACM Comput. Surv. 57, 9, Article 219 (September 2025), 39 pages. <https://doi.org/10.1145/3722214>
- ^{j3} Lewis, N., **Bez, J. L.**, Byna, S., *I/O in Machine Learning Applications on HPC Systems: A 360-degree Survey.* ACM Comput. Surv., 2025, <https://doi.org/10.1145/3722215>

- J4 **Bez, J. L.**, Byna, S., Ibrahim, S. *I/O Access Patterns in HPC Applications: A 360-Degree Survey*. ACM Computing Surveys, 2023, <https://doi.org/10.1145/3611007>
- J5 **Bez, J. L.**, Tang, H., Breitenfeld, S., Zheng, H., Liao, W.-K., Hou, K., Huang, Z., Byna, S. *h5bench: A Unified Benchmark Suite for Evaluating HDF5 I/O Performance on Pre-Exascale Platforms*. Concurrency and Computation: Practice and Experience, 2024, <https://doi.org/10.1002/cpe.8046>
- J6 Carneiro, A. R., **Bez, J. L.**, Osthoff, C., Schnorr, L. M., Navaux, P. O. A. *Uncovering I/O Demands on HPC Platforms: Peeking Under the Hood of Santos Dumont*. Journal of Parallel and Distributed Computing (JPDC), 2022, <https://doi.org/10.1016/j.jpdc.2023.104744>
- J7 **Bez, J. L.**, Zanon Boito, F., Nou, R., Miranda, A., Cortes, T., Navaux, P. O. A. (2020). *Adaptive request scheduling for the I/O forwarding layer using reinforcement learning*. Future Generation Computer Systems, 112, 1156–1169. <https://doi.org/10.1016/j.future.2020.05.005>
- J8 Zaffalon, F., Prisco, A., Souza R., **Bez, J. L.**, Tonin, N., Penna, R. A., Botelho, S. *Estudo Comparativo entre Modelos que Estimam a Habilidade dos Estudantes em Ambientes Virtuais de Programação (Comparative Study between Models that Estimate Student's Skill in Virtual Programming Environments)*. Revista Brasileira de Informática na Educação (Brazilian Journal of Computers in Education), 28, 776–795. <http://dx.doi.org/10.5753/rbie.2020.28.0.776>
- J9 **Bez, J. L.**, Carneiro, A. R., Pavan, P. J., Girelli, V. S., Boito, F. Z., Fagundes, B. A., Osthoff, C., da Silva Dias, P. L., Méhaut, J.-F., Navaux, P. O. (2019). *I/O performance of the Santos Dumont supercomputer*. The International Journal of High Performance Computing Applications, 34(2), 227–245. <https://doi.org/10.1177/1094342019868526>
- J10 Boito, F. Z., Inacio, E. C., **Bez, J. L.**, Navaux, P. O. A., Dantas, M. A. R., Denneulin, Y. (2018). *A Checkpoint of Research on Parallel I/O for High-Performance Computing*. ACM Computing Surveys, 51(2), 1–35. <https://doi.org/10.1145/3152891>
- J11 Pavan, P. J., K. Lorenzoni, R., R. Machado, V., **Bez, J. L.**, Padoin, E. L., Z. Boito, F., Navaux, P. O. A., Méhaut, J. (2018). *Energy efficiency and I/O performance of low-power architectures*. Concurrency and Computation: Practice and Experience, 31(18). <https://doi.org/10.1002/cpe.4948>
- J12 Dagostini, J. I.; **Bez, J. L.**; Tonin, N. A.; Rodegheri, P. R. *Um novo ambiente de discussões para o URI Online Judge*. PERSPECTIVA, v. 41, p. 65-74, 2017.
- J13 **Bez, J. L.**, Bernart, E. E., dos Santos, F. F., Schnorr, L. M., Navaux, P. O. A. (2016). *Performance and energy efficiency analysis of HPC physics simulation applications in a cluster of ARM processors*. Concurrency and Computation: Practice and Experience, 29(22), e4014. <https://doi.org/10.1002/cpe.4014>
- J14 Tonin, N. A.; **Bez, J. L.** *URI Online Judge: A New Interactive Learning Approach*. Computer Technology and Application, v. 4, p. 34-38, 201.

Conferences

- C1 Künas, C., Freytag, G., **Bez, J. L.**, Araújo, T., Navaux, P. *Characterizing Lossless GPU Data Compression Across AMD CDNA and RDNA Architectures*, International Conference on Computational Science and Its Applications (ICCSA), 2026, *Accepted*.
- C2 **Bez, J. L.**, Tang, H., Wang, C., Byna, S., *Data Management in the Continuum: Cross-facility Object-based Data Transfers*, 2025 IEEE 37th International Symposium on Computer Architecture and High Performance Computing (SBAC-PAD), 2025, pp. 46-57, doi: 10.1109/SBAC-PAD66369.2025.00015.
- C3 Egersdoerfer, C., Sareen, A., **Bez, J. L.**, Byna, S., Xu, D. D., and Dai, D., *IOAgent: Democratizing Trustworthy HPC I/O Performance Diagnosis Capability via LLMs*, 2025 IEEE International Parallel and Distributed Processing Symposium (IPDPS), Milano, Italy, 2025, pp. 322-334, doi: 10.1109/IPDPS64566.2025.00036.
- C4 Djebarov, D., Liem, R., Neuwirth, S., **Bez, J. L.**, and Byna, S., *Streamlining HDF5's AI Workloads Benchmarking*, 2025 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW), Milano, Italy, 2025, pp. 722-730, doi: 10.1109/IPDPSW66978.2025.00113.
- C5 Wang, C., Tang, H., **Bez, J. L.**, Byna, S., *Object-Centric Data Management in HPC Workflows - A Case Study*, 2024 IEEE International Conference on Cluster Computing Workshops (CLUSTER Workshops), Kobe, Japan, 2024, pp. 104-108, doi: 10.1109/CLUSTERWorkshops61563.2024.00024.
- C6 Hiniduma, K., Byna, S., **Bez, J. L.**, Madduri, R., *AI Data Readiness Inspector (AIDRIN) for Quantitative Assessment of Data Readiness for AI*, 36th International Conference on Scientific and Statistical Database Management (SSDBM'24). Association for Computing Machinery, New York, NY, USA, Article 7, 1–12. <https://doi.org/10.1145/3676288.3676296>
- C7 Egersdoerfer, C., Sareen, Arnav., **Bez, J. L.**, Byna, S., Dai, D., *ION: Navigating HPC I/O Optimization Journey using Large Language Models*, 16th ACM Workshop on Hot Topics in Storage and File Systems (HotStorage'24), 2024, doi: 10.1145/3655038.3665950

- c8 **Bez, J.L.**, Ather, H., Xia, Y., Byna, S. (2024). *Drilling Down I/O Bottlenecks with Cross-layer I/O Profile Exploration*, 2024 IEEE International Parallel and Distributed Processing Symposium (IPDPS), San Francisco, CA, USA, 2024, pp. 532-543, doi: 10.1109/IPDPS57955.2024.00053
- c9 Rajesh, N., Bateman, K., Byna, B., **Bez, J. L.**, Kougkas, A. (2024). *TunIO: An AI-powered Framework for Optimizing HPC I/O*, 2024 IEEE International Parallel and Distributed Processing Symposium (IPDPS), San Francisco, CA, USA, 2024, pp. 494-505, doi: 10.1109/IPDPS57955.2024.00050
- c10 Ather, H., **Bez, J.L.**, Norris, B., Byna, S. (2023). *Illuminating the I/O Optimization Path of Scientific Applications*. In: Bhatele, A., Hammond, J., Baboulin, M., Kruse, C. (eds) High Performance Computing. ISC High Performance 2023. Lecture Notes in Computer Science, vol 13948. Springer, Cham., doi: 10.1007/978-3-031-32041-5_2
- c11 B. Dong, **J. L. Bez**, S. Byna. *AIIO: Using Artificial Intelligence for Job-level and Automatic I/O Performance Bottleneck Diagnosis*. The 32nd International Symposium on High-Performance Parallel and Distributed Computing (HPDC), (2023), doi: 10.1145/3588195.3592986
- c12 Luettgau, J., Snyder, S., Reddy, T., Awtrey, N., Harms, K., **Bez, J.**, Wang, R., Latham, R., Carns, P., 2023, *Enabling Agile Analysis of I/O Performance Data with PyDarshan*. In Proceedings of the SC '23 Workshops of The International Conference on High Performance Computing, Network, Storage, and Analysis (pp. 1380–1391), <https://doi.org/10.1145/3624062.3624207>
- c13 **J. L. Bez**, A. M. Karimi, A. K. Paul, B. Xie, S. Byna, P. Carns, S. Oral, F. Wang, J. Hanley, *Access Patterns and Performance Behaviors of Multi-layer Supercomputer I/O Subsystems under Production Load*, 31st International ACM Symposium on High-Performance Parallel and Distributed Computing (HPDC '22), 2022, pp. 43-55, doi: 10.1145/3502181.3531461
- c14 T. Li, S. Byna, Q. Koziol, H. Tang, **J. L. Bez**, Q. Kang, *h5bench: HDF5 I/O Kernel Suite for Exercising HPC I/O Patterns*.
- c15 A. R. Carneiro, **J. L. Bez**, C. Osthoff, L. M. Schnorr and P. O. A. Navaux, *HPC Data Storage at a Glance: The Santos Dumont Experience*, 2021 IEEE 33rd International Symposium on Computer Architecture and High Performance Computing (SBAC-PAD), 2021, pp. 157-166, doi: 10.1109/SBAC-PAD53543.2021.00027
- c16 **J. L. Bez**, A. Miranda, R. Nou, F. Z. Boito, T. Cortes and P. Navaux, *Arbitration Policies for On-Demand User-Level I/O Forwarding on HPC Platforms*, 2021 IEEE International Parallel and Distributed Processing Symposium (IPDPS), 2021, pp. 577-586, doi: 10.1109/IPDPS49936.2021.00066.
- c17 **Bez, J. L.**, Boito, F. Z., Nou, R., Miranda, A., Cortes, T., Navaux, P. O. A. (2019, October). *Detecting I/O Access Patterns of HPC Workloads at Runtime*. 2019 31st International Symposium on Computer Architecture and High Performance Computing (SBAC-PAD). <https://doi.org/10.1109/sbac-pad.2019.00025>
- c18 Boito, F. Z., Nou, R., Pilla, L. L., **Bez, J. L.**, Mehaut, J.-F., Cortes, T., Navaux, P. O. A. (2019, July). *On server-side file access pattern matching*. 2019 International Conference on High Performance Computing & Simulation (HPCS). <https://doi.org/10.1109/hpcs48598.2019.9188092>
- c19 Pavan, P. J., **Bez, J. L.**, Serpa, M. S., Boito, F. Z., Navaux, P. O. A. (2019, October). *An Unsupervised Learning Approach for I/O Behavior Characterization*. 2019 31st International Symposium on Computer Architecture and High Performance Computing (SBAC-PAD). <https://doi.org/10.1109/sbac-pad.2019.00019>
- c20 Ramos Carneiro, A., **Bez, J. L.**, Zanon Boito, F., Alves Fagundes, B., Osthoff, C., Navaux, P. O. A. (2018, March). *Collective I/O Performance on the Santos Dumont Supercomputer*. 2018 26th Euromicro International Conference on Parallel, Distributed and Network-Based Processing (PDP). <https://doi.org/10.1109/pdp2018.2018.00015>
- c21 Dagostini, J.; LIMA, M. V. de M.; **Bez, J. L.**; Tonin, N. A. *URI Online Judge Blocks: Construindo Soluções em uma Plataforma Online de Programação*. In: XXIX Simpósio Brasileiro de Informática na Educação (Brazilian Symposium on Computers in Education), 2018, Fortaleza. <http://dx.doi.org/10.5753/cbie.sbie.2018.168>
- c22 Prisco, A., Penna, R., Evandro, Botelho, S., Tonin, N., **Bez, J.** (2018, October). *A multidimensional ELO model for matching learning objects*. 2018 IEEE Frontiers in Education Conference (FIE). <https://doi.org/10.1109/fie.2018.8658847>
- c23 **Bez, J. L.**, Boito, F. Z., Schnorr, L. M., Navaux, P. O. A., Mehaut, J.-F. (2017). *TWINS: Server Access Coordination in the I/O Forwarding Layer*. 2017 25th Euromicro International Conference on Parallel, Distributed and Network-Based Processing (PDP). <https://doi.org/10.1109/pdp.2017.61>
- c24 Boito, F. Z., Bez, J. L., Dupros, F., Dantas, M. A. R., Navaux, P. O. A., Aochi, H. (2017). *High Performance I/O for Seismic Wave Propagation Simulations*. 2017 25th Euromicro International Conference on Parallel, Distributed and Network-Based Processing (PDP). <https://doi.org/10.1109/pdp.2017.92>
- c25 Prisco, A.; Santos, R.; Botelho, S. S. da C.; Tonin, N.; **BEZ, J. L.** *Um Sistema de Recomendação Baseado em um Modelo Cognitivo de Aprendizagem*. In: XXVIII Simpósio Brasileiro de Informática na Educação SBIE (Brazilian Symposium on Computers in Education), 2017, Recife, 2017. v. 1. p. 1667-1676. <http://dx.doi.org/10.5753/cbie.sbie.2017.1667>

- C26 Prisco, A., dos Santos, R., Botelho, S., Tonin, N., **Bez, J.** (2017, October). *Using information technology for personalizing the computer science teaching*. 2017 IEEE Frontiers in Education Conference (FIE). <https://doi.org/10.1109/fie.2017.8190727>
- C27 Dagostini, J.; Lima, M. V. de M.; Bucior, L.; Tonin, N. A.; **Bez, J. L.** *Incentivando a Aprendizagem de Algoritmos Através do URI Online Judge Forum 2.0*. In: XXVIII Simpósio Brasileiro de Informática na Educação SBIE (Brazilian Symposium on Computers in Education), 2017, Recife, 2017. v. 1. p. 1781. <http://dx.doi.org/10.5753/cbie.sbie.2017.1781>
- C28 Pavan, P. J.; Lorenzoni, R. K.; **Bez, J. L.**; Boito, F. Z.; Padoin, E. L.; Navaux, P. O. A.; Mehaut, J. *Eficiência Energética e Desempenho de E/S com Arquiteturas de Baixa Potência*. In: Simpósio em Sistemas Computacionais de Alto Desempenho, 2016, Aracaju. Anais da 17a Edição do Simpósio em Sistemas Computacionais de Alto Desempenho, 2016. v. 1.
- C29 dos Anjos, J. C. S., Assuncao, M. D., **Bez, J.**, Geyer, C., de Freitas, E. P., Carissimi, A., Costa, J. P. C. L., Fedak, G., Freitag, F., Markl, V., Fergus, P., Pereira, R. (2015, October). *SMART: An Application Framework for Real Time Big Data Analysis on Heterogeneous Cloud Environments*. 2015 IEEE International Conference on Computer and Information Technology; Ubiquitous Computing and Communications; Dependable, Autonomic and Secure Computing; Pervasive Intelligence and Computing (CIT/IUCC/DASC/PICOM). <https://doi.org/10.1109/cit/iucc/dasc/picom.2015.29>
- C30 **Bez, J. L.**; Bernart, E. E.; Santos, F. F.; Schnorr, L. M.; Navaux, P. O. A. *Análise da Eficiência Energética de uma Aplicação HPC de Geofísica em um Cluster de Baixo Consumo*. In: WSCAD 2015 - XVI Simpósio em Sistemas Computacionais de Alto Desempenho, v. 1. p. 228-239, Florianópolis, 2015.
- C31 **Bez, J. L.**; Ferreira, C. E.; Tonin, N. A.; *URI Online Judge Academic: A Tool for Professors*. In: 2013 International Conference on Advanced ICT, 2013. <http://dx.doi.org/10.1109/ICCSE.2014.6926445>
- C32 Tonin, N. A.; Zanin, F. A.; **Bez, J. L.** *Enhancing traditional Algorithms classes using URI Online Judge*. In: 2012 International Conference on eLearning and eTechnologies in Education (ICEEE), 2012, Lodz. v. 1. p. 115-118. <http://dx.doi.org/10.1109/ICeLeTE.2012.6333402>
- C33 Tonin, N. A.; Castanho, C. L. O.; **Bez, J. L.** *Using the portal URI Online Judge as a Programming Learning Platform for Computer Science Students*. In: London Internatinal Conference on Education (LICE-2012), v. 1. p. 357-360, 2012.
- C34 Tonin, N. A.; **Bez, J. L.** *URI Online Judge: A New Classroom Tool For Interactive Learning*. In: WORLDCOMP'12 - The 2012 World Congress in Computer Science, Computer Engineering, and Applied Computing, 2012, Las Vegas. FECS 2012. USA: CSREA Press, 2012. v. 1. p. 242-246.

Workshops

- W1 Md K. H. Chowdhury, H. Tang, **J. L. Bez**, P. Bangalore, and S. Byna, *Efficient Asynchronous I/O with Request Merging*, 4th Workshop on Extreme-Scale Storage and Analysis (ESSA), 2023.
- W2 **J. L. Bez**, H. Ather, S. Byna, *Drishiti: Guiding End-Users in the I/O Optimization Journey*, 2022 IEEE/ACM Sixth International Parallel Data Systems Workshop (PDSW), 2022, doi: 10.1109/PDSW56643.2022.00006
- W3 **J. L. Bez**, H. Tang, B. Xie, D. Williams-Young, R. Latham, R. Ross, S. Oral, S. Byna, *I/O Bottleneck Detection and Tuning: Connecting the Dots using Interactive Log Analysis*, 2021 IEEE/ACM Sixth International Parallel Data Systems Workshop (PDSW), 2021, pp. 15-22, doi: 10.1109/PDSW54622.2021.00008
- W4 **Bez, J. L.**, Zanon Boito, F., Nou, R., Miranda, A., Cortes, T., Navaux, P. O. A. *Towards On-Demand I/O Forwarding in HPC Platforms*. International Parallel Data Systems Workshop (PDSW) at The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC20), 2020.
- W5 **Bez, J. L.**, Schnorr, L. M., Navaux, P. O. A. *Characterizing Anomalies of a Multicore ARMv7 Cluster with Parallel N-Body Simulations*. 2015 International Symposium on Computer Architecture and High Performance Computing Workshop (SBAC-PADW). <https://doi.org/10.1109/sbac-padw.2015.18>
- W6 **Bez, J. L.**; Boito, F. Z.; Schnorr, L. M.; Navaux, P. O. A. *Coordinating Data Access at I/O Forwarding Nodes*. In: Workshop de Processamento Paralelo e Distribuído. Porto Alegre, 2016. v. 1. p. 9-11.
- W7 Machado, V. R.; Boito, F. Z.; Kassick, R. V.; **Bez, J. L.**; Navaux, P. O. A.; Denneulin, Y. *Parallel Storage Devices Profiling with SeRRa*. In: 14º WPerformance - XIV Workshop em Desempenho de Sistemas Computacionais e de Comunicação, 2015, Recife.
- W8 Selivon, M.; **Bez, J. L.**; Tonin, N. A. *URI Online Judge Academic: Integração e Consolidação da Ferramenta no Processo de Ensino/Aprendizagem*. In: 23º WEI - Workshop sobre Educação em Computação, 2015, Recife. Anais / XXXV Congresso da Sociedade Brasileira de Computação, 2015.

Short Papers

- S1 Lima, M. V. M.; Rodegheri, P. R.; Tonin, N. A.; **Bez, J. L.** *Uma Ferramenta Online para Execução de Scripts em SQL*. In: Escola Regional de Banco de Dados, 2017, Passo Fundo. Anais da XIII Escola Regional de Banco de Dados, 2017. Passo Fundo: Ed. Universidade de Passo Fundo, 2017. v. 1. p. 127-130.

- S2 Rampon, N. G.; Boito, F. Z.; **Bez, J. L.**; Navaux, P. O. A. *Otimização de E/S Paralela em Simulações de Ondas Sísmicas*. In: Escola Regional de Alto Desempenho (ERAD/RS), 2017, Ijuí. Anais da XVI Escola Regional de Alto Desempenho (ERAD/RS), 2017. v. 1. p. 343-346.
- S3 Machado, V. R.; Braga, A. B.; Rampon, N. G.; Boito, F. Z.; **Bez, J. L.**; Padoin, E. L.; Navaux, P. O. A. *Avaliação do Consumo Energético de processadores ARM em Sistemas de Arquivos Paralelos*. In: Escola Regional de Alto Desempenho (ERAD/RS), 2017, Ijuí. Anais da XVII Escola Regional de Alto Desempenho (ERAD/RS). Ijuí: Impressos Portão, 2017. v. 1. p. 263-266.
- S4 Braga, A. B.; Machado V. R.; **Bez, J. L.**; Boito, F. Z.; Navaux, P. O. A. *Influência do Padrão de Acesso na Eficiência Energética de Dispositivos SSDs*. In: Escola Regional de Alto Desempenho (ERAD/RS), 2017, Ijuí. Anais da XVII Escola Regional de Alto Desempenho (ERAD/RS). Ijuí: Impressos Portão, 2017. v. 1. p. 331-334.
- S5 **Bez, J. L.**; Boito, F. Z.; Schnorr, L. M.; Navaux, P. O. A. *Escalonamento de I/O em Servidores de Encaminhamento*. In: Escola Regional de Alto Desempenho (ERAD/RS), 2016, São Leopoldo. Anais da XVI Escola Regional de Alto Desempenho (ERAD/RS). São Leopoldo: Impressos Portão, 2016. v. 1. p. 173-174.
- S6 Pavan, P. J.; Lorenzoni, R. K.; **Bez, J. L.**; Boito, F. Z.; Padoin, E. L.; Navaux, P. O. A. . *Análise de Consumo Energético e Desempenho de Operações E/S em Arquiteturas de Baixa Potência*. In: Workshop de Processamento Paralelo e Distribuído. Porto Alegre, 2016. v. 1. p. 5-9.
- S7 Braga, A. B.; Rampon, N. G.; Machado, V. R.; **Bez, J. L.**; Boito, F. Z.; Kassick, R. V.; Padoin, E. L.; Navaux, P. O. A. . *Viability of Low-Power Architectures as Parallel File Systems*. In: Workshop de Processamento Paralelo e Distribuído. Porto Alegre, 2016. v. 1. p. 27-30.
- S8 **Bez, J. L.**; Boito, F. Z.; Kassick, R. V.; Machado, V. R.; Navaux, P. O. A. . *Faster Storage Devices Profiling with Parallel SeRRa*. In: Workshop de Processamento Paralelo e Distribuído, 2015, Porto Alegre. v. 1. p. 33-36.

Posters

- P1 Hiniduma, K., **Bez, J. L.**, Madduri, R., Byna, S., *AIDRIN: A Comprehensive Toolset for Automating Data Preparation for AI*. SC25, 2025.
- P2 Yazdani, A., **J. L. Bez**, Paul, A. K., Karimi, A. M., Byna, S., Butt, A., *Characterization of the Concurrent Application I/O Interference in Leadership Scale Systems: A Focus on I/O Optimization*, 38th IEEE International Parallel and Distributed Processing Symposium (IPDPS'24), 2024.
- P3 **J. L. Bez**, Ather, H., Byna, S., *Drishti: Connecting Metrics to I/O Performance Improvements*, Monterey Data Conference (MDC' 23), 2023.
- P4 **J. L. Bez**, Ather, H., Byna, S., *Drishti: Connecting Metrics to I/O Performance Improvements*, Monterey Data Conference (MDC' 23), 2023.
- P5 **J. L. Bez**, A. M. Karimi, A. K. Paul, B. Xie, S. Byna, P. Carns, S. Oral, F. Wang, J. Hanley, *Access Patterns and Performance Behaviors of Multi-layer Supercomputer I/O Subsystems under Production Load*, 31st International ACM Symposium on High-Performance Parallel and Distributed Computing (HPDC '22), 2022.
- P6 **Bez, J. L.**, Boito, F. Z., Nou, R., Miranda, A., Cortes, T., Navaux, P. O. A. (2019, October). *Towards the Reconfiguration of the I/O Forwarding Layer*. 35th IEEE International Parallel & Distributed Processing Symposium.

Book Chapters

- B1 Serpa, M. S.; **Bez, J. L.**; Cruz, E. H. M.; Diener, M.; Zanata, M. A.; Navaux, P. O. A. . *Intel Modern Code: Programação Vetorial e Paralela em Arquiteturas Intel Xeon e Intel Xeon Phi*. In: Edson Luiz Padoin; Márcia Cristina Cera; Andrea Schwertner Charão. (Org.). Escola Regional de Alto Desempenho. 1ed.Ijuí: , 2017, v. 1, p. 63-80.

Media

- M1 **Bez, J. L.** (2018). *Computer Scientists in Action: Jean Luca Bez, High Performance Computing*. XRDS: Crossroads, The ACM Magazine for Students, 25(1), 64–65. <https://doi.org/10.1145/3265939>.

PRESENTATIONS

Panel

- P1 **Bez, J. L.**, Klymko, K., de Jong, B. (2025). *CSASP Panel – Career Paths: How did I get here?*. CS Summer Student Program 2025 Career Panel held by LBNL.

- P2 Dufek, A., Nonaka, A., Trebotich, D., **Bez, J. L.** (2023). *DOE Researcher Panel: How HPC can advance discovery*. Introduction to High-Performance Computing Energy Justice Bootcamp, held by ALCF, ECP, NERSC, OLCF, and Sustainable Horizons Institute (SHI).

Research Talks

- RT1 **Bez, J. L.**, Byna, S., Ather, H., “‘*Drishti and HDF5: What is actually happening in my application?*’, HDF5 User Group (HUG) Meeting, 2023.
- RT2 **Bez, J. L.**, “‘*Where’s the bottleneck?*’, Berkeley Lab Research SLAM, 2022.
- RT3 **Bez, J. L.**, *Towards Understanding I/O Behavior with Interactive Exploration*, Lawrence Berkeley National Laboratory’s Computing Sciences Area 2022 Postdoc Symposium, Berkeley, USA, 2022.
- RT4 **Bez, J. L.**, Byna, S. *Visualizing Darshan Extended Traces*. Analyzing Parallel I/O – Birds of a Feather, SC’21: Supercomputing, St. Louis, USA, 2021.
- RT5 **Bez, J. L.**, Boito, F. Z., Nou, R., Miranda, A., Cortes, T., Navaux, P. O. A. *A Reinforcement Learning Strategy to Tune Request Scheduling at the I/O Forwarding Layer*. HPC-IODC: HPC I/O in the Data Center Workshop, ISC High Performance (Virtual), 2020.
- RT6 **Bez, J. L.**, Boito, F. Z., Nou, R., Miranda, A., Cortes, T., Navaux, P. O. A. *Dynamic Reconfiguration of the I/O Forwarding Layer of HPC Architectures*. JLESC – Joint Laboratory on Extreme Scale Computing, Barcelona, Spain, 2018.

Invited Talks

- IT1 **Bez, J. L.**, *Visualizing I/O Bottlenecks with DXT Explorer 2.0*, Analyzing Parallel I/O – Birds of a Feather, SC’22: Supercomputing, Dallas, USA, 2022.
- IT2 Chiusole, A. **Bez, J. L.** *I/O Profiling on Perlmutter with Darshan*, NERSC Data Day, Berkeley, USA, 2022.
- IT3 **Bez, J. L.**, Byna, S. *Understanding I/O Behavior with Interactive Darshan Log Analysis*. ECP Community Days BoF, Online, USA, 2022.
- IT4 **Bez, J. L.**, Byna, S. *Visualizing Darshan Extended Traces*, Analyzing Parallel I/O – Birds of a Feather, SC’21: Supercomputing, St. Louis, USA, 2021.
- IT5 **Bez, J. L.**, Byna, S. *Towards Understanding I/O Behavior with Interactive Exploration*, SciData – Division Meeting Talk, Lawrence Berkeley National Laboratory, Berkeley, USA, 2022.
- IT6 **Bez, J. L.**, Byna, S. *Dynamic Tuning and Reconfiguration of the I/O Forwarding Layer in HPC Platforms – Ph.D. Summary*, SciData – Division Meeting Talk, Lawrence Berkeley National Laboratory, Berkeley, USA, 2021.
- IT7 **Bez, J. L.** *De Arquivos a Big Data: Como Supercomputadores Armazenam Dados*. Semana Acadêmica Integrada dos Cursos de Ciência da Computação da URI, Brazil, 2020.
- IT8 **Bez, J. L.**; Tonin, N. A. *URI Online Judge: Plataforma e Desafios*. Sétima Semana Acadêmica Integrada das Ciências Computacionais – VII SAICC, Universidade Federal do Rio Grande (FURG), Rio Grande, Brazil, 2017.
- IT9 Tonin, N. A. ; **Bez, J. L.** *URI Online Judge - Programação, Aprendizado e Desafios*. Serviço Nacional de Aprendizagem Comercial (SENAC), Passo Fundo, Brazil, 2012.
- IT10 Tonin, N. A. ; **Bez, J. L.** *Portal URI Online Judge e a Maratona de Programação*. I Workshop Desafio Pela Inovação. Universidade de Passo Fundo, Passo Fundo, Brazil, 2012.
- IT11 Tonin, N. A. ; **Bez, J. L.** *URI Online Judge, um portal para a Prática de Programação*. XXI Simpósio de Inovação em Tecnologias Computacionais. Universidade Regional Integrada do Alto Uruguai e das Missões, Santo Ângelo, Brazil, 2012 (*Invited Talk*).
- IT12 Tonin, N. A. ; **Bez, J. L.** *URI Online Judge - Programação, Aprendizado e Desafios*. Universidade Regional Integrada do Alto Uruguai e das Missões, Frederico Westphalen, Brazil, 2012.
- IT13 Tonin, N. A. ; **Bez, J. L.** *URI Online Judge, um Portal para a Prática de Programação*. SITIS 2012 - XXI Simpósio de Inovação em Tecnologias Computacionais. Universidade Regional Integrada do Alto Uruguai e das Missões, Santo Ângelo, Brazil, 2012.

Others

- P1 **Bez, J. L.**; Navaux, P. O. A. *Evaluating I/O Scheduling Techniques at the Forwarding Layer and Coordinating Data Server Accesses*. 31º Concurso de Teses e Dissertações (CTD) – XXXVIII Congresso da Sociedade Brasileira de Computação. Sociedade Brasileira de Computação. Natal, Rio Grande do Norte, Brazil, 2018.

- P2 **Bez, J. L.**; Tonin, N. A. *URI Online Judge: Coding and Beyond*. ACM International Collegiate Programming Contest (ICPC) Collaborative Learning Institute Symposium (CLIS), Phuket, Thailand, 2016.
- P3 **Bez, J. L.**; Boito, F. Z.; Schnorr, L. M.; Navaux, P. O. A. . *Coordinating Data Access at I/O Forwarding Nodes*. Workshop de Processamento Paralelo e Distribuído. Porto Alegre, Brazil, 2016.
- P4 **Bez, J. L.**, Schnorr, L. M., Navaux, P. O. A. *Characterizing Anomalies of a Multicore ARMv7 Cluster with Parallel N-Body Simulations*. 2015 International Symposium on Computer Architecture and High Performance Computing Workshop (SBAC-PADW), Florianópolis, Brazil, 2016.
- P5 **Bez, J. L.**; Tonin, N. A.; Rodegheri, P. R. *URI Online Judge Academic: A Tool for Algorithms and Programming Classes*. IEEE International Conference on Computer Science & Education. Univ. of British Columbia, Vancouver, Canada, 2014.
- P6 **Bez, J. L.**; Tonin, N. A. *The URI Online Judge Learning Platform*. ACM International Collegiate Programming Contest (ICPC) Collaborative Learning Institute Symposium (CLIS), Sanit Petersburg, Russia, 2013.
- P7 Tonin, N. A. ; **Bez, J. L.** *URI Online Judge: A New Classroom Tool For Interactive Learning*. WORLDCOMP'12 - The 2012 World Congress in Computer Science, Computer Engineering, and Applied Computing, Las Vegas, USA, 2012.

GRANTS AND FELLOWSHIPS

Grants

- G1 **Lawrence Berkeley National Laboratory (LBNL)**. Laboratory Directed Research and Development (LDRD) Early Career. *Enhancing AI Data Readiness in Scientific Data: Integration, Automation, and Human-in-the-Loop Approaches*. Principal Investigator (PI), 2025 – 2027.
- G2 **Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES)**. Programa Institucional de Internacionalização (PRINT). "Sandwich" Doctorate / Doctoral Stay. Grant 88887.363381/2019-00 (€ 9,640.00).

Scholarships

- S1 **Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq)**. *Ph.D. Scholarship* (2017 – 2021).
- S2 **Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq)**. *Masters Scholarship* (2015).

AWARDS

Computing's Top 30 Early Career Professionals for 2025 – *January 2026*

IEEE Computer Society

SBAC-PAD 2025 Paper Recognition Award – *October 2025*

"Data Management in the Continuum: Cross-facility Object-based Data Transfers"

37th IEEE/SBC International Symposium on Computer Architecture and High Performance Computing (SBAC-PAD)

LBNL IPO Office – Inventor of the Year Award – *November 2023*

Lawrence Berkeley National Laboratory

Better Scientific Software (BSSw) Fellow

Honorable Mention — *January 2023*

Better Scientific Software – bssw.io

Berkeley Lab Research SLAM Finalist

Finalist (Computer Sciences) — *September 2022*

"Where's the bottleneck?"

Lawrence Berkeley National Laboratory

Contest of Thesis and Dissertations in Computer Architecture and High-Performance Computing

First Place (Thesis) — *October 2021*

"Dynamic Tuning and Reconfiguration of the I/O Forwarding Layer in HPC Platforms"

Simpósio de Sistemas Computacionais de Alto Desempenho, promoted by the Brazilian Computer Society

Contest of Thesis and Dissertations in Computer Architecture and High-Performance Computing

First Place (Dissertations) — *October 2017*

"Evaluating I/O Scheduling Techniques at the Forwarding Layer and Coordinating Data Server Accesses"
Simpósio de Sistemas Computacionais de Alto Desempenho, promoted by the Brazilian Computer Society

Brazilian Computer Society (SBC) Highlight Student Award (*Summa Cum Laude* – 9.517 / 10.0)
Awarded by Brazilian Computer Society (SBC) – February 2015

URI Erechim Best Student Award (*Summa Cum Laude* – 9.517 / 10.0)
Awarded by Universidade Regional Integrada do Alto Uruguai e das Missões (URI) – February 2015

URI University Extension Projects — Highlight Student Award*
Awarded by Universidade Regional Integrada do Alto Uruguai e das Missões (URI) – 2013
URI Online Judge Project (urionlinejudge.edu.br)

URI University Extension Projects — Highlight Student Award*
Awarded by Universidade Regional Integrada do Alto Uruguai e das Missões (URI) – 2012
URI Online Judge Project (urionlinejudge.edu.br)

* Awarded to undergraduate students for research and extension projects based on results and impact on the community.

TEACHING EXPERIENCE

Teaching Assistant

TA1 *Advanced Computer Architectures* (2016/1)
Federal University of Rio Grande do Sul
Advisor: Prof. Dr. Philippe O. A. Navaux (UFRGS)

TA2 *Compilers* (2015/2)
Federal University of Rio Grande do Sul
Advisor: Prof. Dr. Philippe O. A. Navaux (UFRGS)

Tutorials and Short Term Courses

- ST1 **Bez, J. L.**. *Introdução a Programação MPI com Extensões para E/S (MPI-IO)* ("Introduction to MPI Programming with I/O Extensions (MPI-IO)"). Escola de Verão Supercomputador SDumont. National Laboratory for Scientific Computing (LNCC), Petrópolis, Brazil, 2019.
- ST2 **Bez, J. L.**. *Introdução ao MPI-IO* ("Introduction to MPI-IO"). Escola Regional de Alto Desempenho do Estado do Rio Grande do Sul (ERAD/RS 2018), Porto Alegre, Brazil, 2018.
- ST3 **Bez, J. L.**; Boito, F. Z. ; Navaux, P. O. A. . *Introdução a Programação MPI com Extensões para E/S (MPI-IO)* ("Introduction to MPI Programming with I/O Extensions (MPI-IO)"). Escola Supercomputador SDumont. National Laboratory for Scientific Computing (LNCC), Petrópolis, Brazil, 2018.
- ST4 Alves, M. A. Z.; Diener, M.; Cruz, E. H. M.; **Bez, J. L.**; Navaux, P. O. A. *Workshop Parallel Programming and Optimization for Intel Architecture*. UniRitter, Porto Alegre, Brazil, 2016.
- ST5 **Bez, J. L.**; Nava, M. L. *Desenvolvimento Web* ("Web Development"). Universidade Regional do Alto Uruguai e das Missões, Erechim, 2014.
- ST6 **Bez, J. L.**; Nava, M. L. *Desenvolvimento Web* ("Web Development"). Universidade Regional do Alto Uruguai e das Missões, Erechim, 2013.
- ST7 Tonin, N. A.; **Bez, J. L.** *Informática para Crianças de Ensino Fundamental no Programa AABB Comunidade* ("Computer Science for Elementary School Children in the AABB Community Program"). Universidade Regional do Alto Uruguai e das Missões, Erechim, Brazil, 2011.

ADVISING

Intern Mentoring

- IM1 Kaveen Hiniduma, Graduate Student Researcher, Lawrence Berkeley National Laboratory, 2026.
- IM2 Harish Balaji, Summer Intern, Google Summer of Code (GSoC 2025), Lawrence Berkeley National Laboratory, 2025.

- IM3 Henry Zhou, Summer Intern, Google Summer of Code (GSoC 2024), Lawrence Berkeley National Laboratory, 2024.
- IM4 Joel Tony, Summer Intern, Google Summer of Code (GSoC 2024), Lawrence Berkeley National Laboratory, 2024.
- IM5 Ahmad Hossein Yazdani, Summer Intern, Lawrence Berkeley National Laboratory, 2023.
- IM6 Hammad Ather, Summer Intern / Intern, Lawrence Berkeley National Laboratory, 2022 – 2023.
- IM7 Asses Kaur, Summer Intern, Lawrence Berkeley National Laboratory, 2022.
- IM8 Neeraj Rajesh, Summer Intern, Lawrence Berkeley National Laboratory, 2021.
- IM9 Kenneth Casimiro, Summer Intern, Lawrence Berkeley National Laboratory, 2021.

Undergraduate Thesis

- UT1 Matheus Felipe Fabian. *Aprimoramento das Competições na Plataforma URI Online Judge*. 2021.
Universidade Regional Integrada do Alto Uruguai e das Missões
Advisor: Neilor A. Tonin (URI)
Co-Advisor: **Jean Luca Bez** (UFRGS)
- UT2 Jessica Imlau Dagostini. *Avaliação da Aplicabilidade de Containers em um Sistema Distribuído de Julgamento de Códigos*. 2019
Universidade Regional Integrada do Alto Uruguai e das Missões
Advisor: Neilor A. Tonin (URI)
Co-Advisor: **Jean Luca Bez** (UFRGS)

RESEARCH PROJECTS

- RP1 **Enhancing AI Data Readiness in Scientific Data: Integration, Automation, and Human-in-the-Loop Approaches** 2025 – Present
PI for this LDRD (Laboratory Directed Research and Development) research project. We propose exploring and designing standardized metrics to evaluate the AI readiness of datasets, recommending and implementing remedies to enhance data readiness. Focusing initially on High Energy Physics (HEP) data and later expanding to other datasets and domains, the project aims to build a multi-domain solution for reading data for AI-enabled scientific exploration and discovery.
- RP2 **Transformational AI Models Consortium (ModCon) (DOE Genesis Mission)** 2025 – Present
Member of the LBNL team in ModCon, a DOE ASCR consortium under the Genesis Mission building transformational AI models for science. I contribute to both the Data Brokers & Standards (DBS) and BASE Data teams, bridging the two, and co-lead the DBS effort to automate data curation. My work focuses on AI data readiness for scientific datasets, including the design of data-card and tool-card standards and generator tooling, the growth and curation of the shared data/tool catalog, and provenance and observability for agentic data-preparation pipelines.
- RP3 **Exploring the Power of Distributed Intelligence for Resilient Scientific Workflows (SWARM)** 2025 – Present
Institutional PI (LBNL) for SWARM, a DOE-funded effort whose goal is to design Swarm Intelligence (SI)-based resilient Integrated Research Infrastructure (IRI) that can quickly recover from failures, adapt to changes in the environment, and maximize resource utilization. My contribution centers on the data-management aspect: designing the distributed, agent-based services that coordinate data placement and movement across facilities using live network and facility metrics, and validating their resilience through fault- and failure-injection testing. As Institutional PI, I lead LBNL's data-management contributions to the project.
- RP4 **SciDAC Institute for Artificial Intelligence, Computer Science, and Data (RAPIDS3)** 2025 – Present
Collaborator on RAPIDS3, a DOE ASCR SciDAC Institute providing computer-science and data expertise to DOE science teams. I contribute to the data-management thrust, supporting HDF5 stewardship and I/O characterization and bringing tools such as h5bench, Drishti, and DXT-Explorer to bear on facility and application workloads.
- RP5 **End-to-end Object-focused Data Management Runtime System** 2023 – 2025
Collaborator on the E2E project, which designs an end-to-end, object-focused data management runtime spanning the full life cycle of scientific data objects (from compute-node memory and facility file systems and tape archives to remote facilities and cloud storage) with the resulting techniques integrated into the Proactive Data Containers (PDC) framework. My contributions include the end-to-end object transfer layer enabling multiple storage backends (e.g., S3 and VAST) in PDC and PDC's continuous integration and performance-regression testing.

RP6 Proactive Data Containers (PDC)

2021 – 2023

Collaborator on the PDC project that seeks to provide a novel data abstraction for managing science data in an object-oriented manner. PDC covers multiple R&D efforts such as formulation of object-oriented PDCs and their mapping in different levels of the exascale storage hierarchy, efficient strategies for moving data in deep storage hierarchies, techniques for transforming and reorganizing data based on application requirements, and novel analysis paradigms for enabling data transformations and user-defined analysis on data in PDCs.

RP7 Exascale Computing Project (ECP) (exascaleproject.org)

2021 – 2023

Key member on the ECP's ExaIO team that develops HDF5 and UnifyFS libraries in the Exascale Computing Project. Collaborator on the ECP ST/App Integration. ECP is an effort of the US Department of Energy (DOE). ECP aims to deliver breakthrough modeling and simulation solutions that analyze more data in less time, providing insights and answers to the most critical US challenges in scientific discovery, energy assurance, economic competitiveness, and national security. This project encompasses applications, system software, hardware technologies and architectures, and workforce development to meet the scientific and national security mission needs of DOE.

RP8 Towards Exascale I/O

2018 – 2021

Collaborator on the Towards Exascale I/O research project that combines the skills of the Parallel and Distributed Processing Group (GPPD) of the Federal University of Rio Grande do Sul with the research teams of the Barcelona Supercomputing Center (BSC), INRIA, and the National Laboratory for Scientific Computing (LNCC). The GPPD has maintained a line of research on HPC storage for more than 15 years, through various postgraduate studies that have resulted in indexed publications, in addition to several doctoral theses and master's dissertations in I/O for HPC. The BSC Storage group has recognized experience in I/O for HPC due to its expertise with the MareNostrum supercomputer. INRIA Grenoble and Bordeaux have recognized experience in projects with high storage capacities and I/O. Finally, this project aims to bring collaborations in parallel I/O for HPC by proposing and evaluating strategies that help reach Exascale.

Financed by the Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq), Brazil.

RP9 High Performance Computing for Energy (HPC4E) (hpc4e.eu)

2016 – 2016

Collaborator on the HPC4E project, which aimed to apply new exascale HPC techniques to energy industry simulations, customizing them, and going beyond the state-of-the-art for different energy source simulations. The HPC4E consortium encompassed 13 partners, seven from Europe, and six from Brazil.

COMMUNITY SERVICE

Community Service

CS1 NERSC User Group Executive Committee (NUGEX), NERSC, 2022 – Present

CS2 ACM SIGHPC: Membership Coordinator, 2021 – Present

CS3 ACM SIGHPC Computing Continuum Chapter: Vice-Chair, 2025 – Present

CS4 ACM SIGHPC Computing Continuum Chapter: Secretary, 2023 – 2025

CS5 ACM SIGHPC Computing Continuum Chapter: Founder

CS6 IO500 Committee Member: 2024 – Present

CS7 IO500: Web Designer, 2020 – Present

Editorial Board

EB1 Future Generation Computer Systems (FGCS): Guest Editor for Special Issue on Cloud Continuum, 2025

EB2 Frontiers in High Performance Computing: Review Editor for HPC Applications, 2023 – Present

EB3 Frontiers in High Performance Computing: Review Editor for High Performance Big Data Systems, 2023 – Present

Program Committee

PC1 SC'27: Workshop and Symposia Co-Chair, 2027

PC2 IPDPS'27: System Software Track Chair, 2027

PC3 HPCAsia'27: Programming Models and System Software PC Member, 2027

PC4 SC'26: Reproducibility Initiative Chair, 2026

PC5 HPDC'26: HPDC Workshops Co-chair, 2026

PC6 SBAC-PAD'26: Distributed Systems, Networking, and Storage Track Co-chair, 2026

PC7 ICPP'26: Demo Co-chair, 2026

PC8 ICPP'26: Performance Track PC Member, 2026

PC9 ESSA'26: PC Member, 2026

PC10 SSDBM'26: Poster Committee Member, 2026

PC11 CLUSTER'26: Poster Committee Member, 2026

PC12 CARLA'26: Workshop on Disruptive Continuum Computing for a Sustainable Future, Chair, 2026

PC13 CARLA'26: PC Member, 2026

PC14 CHEOPS'26: Challenges and Opportunities of Efficient and Performant Storage Systems, PC member, 2026

PC15 Berkeley Lab Research SLAM: Screening Judge, Lawrence Berkeley National Laboratory, 2026

PC16 Berkeley Lab Research SLAM: Screening Judge, Lawrence Berkeley National Laboratory, 2025

PC17 DRAI'25: Data Readiness for AI Workshop Co-Chair, 2025

PC18 WISDOM'25: Workflows, Intelligent Scientific Data and Optimization for Automated Management Co-Chair, 2025

PC19 SC'25 IO500: Birds-of-a-Feather Organizer, 2025

PC20 SC'25: New Volunteers Chair, Inclusivity, 2025

PC21 SC'25: Finance Liaison for Technical Program, Financial Management, 2025

PC22 SSDBM'25: Proceedings Chair, 2025

PC23 SBAC-PAD'25: Publicity Chair, 2025

PC24 CCGrid'25: SCALE Challenge Track Co-chair, 2025

PC25 SC'24: Reproducibility Challenge Co-chair, 2024

PC26 CARLA'24: Latin America High Performance Computing Conference, 2024

PC27 PDSW'24: Reproducibility Co-chair, 2024

PC28 SBAC-PAD'24: Workshop Proceedings Chair, 2024

PC29 SBAC-PAD'24: Workshop Co-chair, 2024

PC30 SSDBM'24: Proceedings Chair, 2024

PC31 CCGrid'24: SCALE Challenge Track Co-chair, 2024

PC32 HPDC'24: Technical Program Committee Member, 2024

PC33 HiPC'23: Student Research Symposium Committee Member, 2023

PC34 SC'23: Reproducibility Challenge Committee Member, 2023

PC35 SC'23: Artifact Description/Article Evaluation (AD/AE) Submissions Committee Member, 2023

PC36 PDSW'23: Technical Program Committee Member, 2023

PC37 CCGrid'23: PC member of Research Papers (Artifact Evaluation) and SCALE Challenge, 2023

PC38 FIE'23: IEEE Frontiers in Education Conference, Reviewer, 2023

PC39 Escola Regional de Alto Desempenho da Região Sul – ERAD/RS – Fórum de Pós Graduação, 2023

PC40 CHEOPS'23: Challenges and Opportunities of Efficient and Performant Storage Systems, PC member, 2023

PC41 SSDBM'23: Scientific and Statistical Database Management Conference, PC member, 2022

PC42 Cluster'23: Programming and System Software Papers Committee Member, 2023

PC43 PDSW'22: Committee Member (Publicity Chair), 2022

- PC44 HPCC'22: Technical Program Committee Member, 2022
- PC45 SC'22: Students@SC, PC member, 2022
- PC46 SC'22: Artifact Description/Article Evaluation (AD/AE) Submissions Committee Member, 2022
- PC47 SSDBM'22: Scientific and Statistical Database Management Conference, PC member, 2022
- PC48 REX-IO'22: Workshop on Re-envisioning Extreme-Scale I/O for Emerging Hybrid HPC Workloads, PC member, 2022
- PC49 CCGrid'22: PC member of Storage and I/O Systems, 2022
- PC50 Escola Regional de Alto Desempenho da Região Sul – ERAD/RS – Fórum de Iniciação Científica, 2022
- PC51 SC'21: Student Volunteers (Student Volunteer Reviewers, Virtual Logistics Liaison), 2021
- PC52 SC'21: Virtual Logistics Operations Subcommittee (Virtual Logistics Student Volunteer Liaison), 2021
- PC53 SC'21: Artifact Description/Article Evaluation (AD/AE) Submissions Committee Member, 2021
- PC54 REX-IO'21: Workshop on Re-envisioning Extreme-Scale I/O for Emerging Hybrid HPC Workloads, PC member, 2021
- PC55 Escola Regional de Alto Desempenho da Região Sul – ERAD/RS – Fórum de Iniciação Científica, 2021
- PC56 WSPPD'20: XV Workshop de Processamento Paralelo e Distribuído/UFRGS, PC member, 2020
- PC57 Escola Regional de Alto Desempenho da Região Sul – ERAD/RS – Fórum de Iniciação Científica, 2020
- PC58 WSPPD'19: XV Workshop de Processamento Paralelo e Distribuído/UFRGS, PC member, 2019
- PC59 WSPPD'18: XV Workshop de Processamento Paralelo e Distribuído/UFRGS, PC member, 2018
- PC60 WSPPD'17: XV Workshop de Processamento Paralelo e Distribuído/UFRGS, PC member, 2017
- PC61 WSPPD'16: XV Workshop de Processamento Paralelo e Distribuído/UFRGS, PC member, 2016
- PC62 WSPPD'15: XV Workshop de Processamento Paralelo e Distribuído/UFRGS, PC member, 2015

External Reviewer

- ER1 SC'23: International Conference for High Performance Computing, Networking, Storage, and Analysis, 2023
- ER2 IEEE Internet Computing, 2023
- ER3 Cluster'22: Programming and System Software Track, 2022
- ER4 ICPP'22: 51st International Conference on Parallel Processing, 2022
- ER5 Cluster'21: Data, Storage and Visualization / Programming and System Software Track, 2022
- ER6 CCGrid'21: 21st IEEE/ACM international Symposium on Cluster, Cloud and Internet Computing, 2021
- ER7 Euro-Par'21: 27th International Conference on Parallel and Distributed Computing, 2021
- ER8 ICS'20: 34th ACM International Conference on Supercomputing, 2020
- ER9 UCC'18: 11th IEEE/ACM International Conference on Utility and Cloud Computing, 2018

Event Organization

- E01 Navaux, P. O. A., Geyer, C., Schnorr, L. M., **BEZ, J. L.**, Serpa, M. S., Souza Junior, P., *XV Workshop de Processamento Paralelo e Distribuído/UFRGS - WSPPD 2017*, Universidade Federal do Rio Grande do Sul, Instituto de Informática, Porto Alegre, 2017.

Outreach Projects

- OP1 **URI Online Judge / beecrowd** (urionlinejudge.com.br / beecrowd.com.br) 2012 – Present
 URI Online Judge is an educational project created in Brazil by Neilor Tonin and Jean Luca Bez to help students learn and practice algorithms and programming languages. It has more than 500,000 registered users worldwide and has automatically evaluated over 20 million source-code solutions for programming assignments. The platform has received more than 20 million solutions. It has also received a Brazilian National Award (Prêmio Santander Universidades) for its impact and importance. The project was presented at the ACM International Collegiate Programming Contest (ICPC) World Finals in St. Petersburg, Russia, and in Phuket, Thailand.

VOLUNTEER EXPERIENCE

- VE1 **Selected Lead Student Volunteer** at ACM/IEEE SC'20 Supercomputing Conference, Atlanta (USA), 2020.
Guided Interest Groups (GIG) on Storage and I/O for HPC
- VE2 **Selected Student Volunteer** at ACM/IEEE SC'19 Supercomputing Conference, Denver (USA), 2019.
- VE3 **ICPC Judge** at the International Collegiate Programming Contest (ICPC) South America/Brazil First Phase, 2018.
- VE4 **ICPC Deputy/Student Volunteer** at International Collegiate Programming Contest (ICPC), Phuket, Thailand, 2016.

LANGUAGES

- L1 Portuguese (Native)
- L2 English (Fluent – C1)
- L3 French (Basic – A2.1)
- L4 Catalan (Basic – A2.1)